

Signals and Systems time domain representation and Signal Operations

Participated in & did the coursework [Y/N]?

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1 INTRODUCTION

1.1 Abstract

This report summarizes Laboratory Exercise 2, which focuses on the time-domain representation of signals and their corresponding operations. The primary objectives include determining symbolic mathematical expressions for continuous-time signals and utilizing MATLAB to sketch these sequences accurately. Essential signal operations such as time shifting, time reversal, and time compression are applied to a base signal to visually demonstrate their theoretical effects in the time domain. Furthermore, the coursework explores the mathematical formulation and plotting of impulse responses for standard filters, specifically low-pass, high-pass, bandpass, and bandstop filters. Finally, the auto-correlation of a specified signal is computed and graphed against its lag time to analyze its self-similarity.

1.2 Conspectus

1.3 What are the objectives of the coursework?

- 1) To write symbolic math expressions for continuous-time signals and specific filter impulse responses;
- 2) To successfully sketch multiple sequences demonstrating time-domain operations like shifting, flipping, and compression;

1.4 How does the coursework fit the course and previously done coursework?

By:

- 1) Applying practical MATLAB programming to execute theoretical signal operations covered in the Signals and Systems course.
- 2) Translating symbolic math expressions into visual time-domain representations for deeper behavioral analysis.

By affixing my/our signature/s, I/we, the author/s, pledge that: I/we have completed this coursework on my/our own; I/we have not used any unauthorized material/assistance/help on this coursework; and I/we have not given directly or indirectly to any other student/unauthorized person/means any access to any part of the specified coursework. Coram Deo.

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1.5 How were the objectives achieved?

By:

- 1) Utilizing MATLAB functions such as heaviside, dirac, and exp to symbolically define the input signals and filter responses.
- 2) Using plotting commands like subplot to arrange multiple graphs into a single figure and xcorr to determine the auto-correlation.

1.6 What are the key results and generalizations?

The key results are:

- 1) Time-domain operations were visually verified, showing accurate signal shifts, reversals, and scaling based on their mathematical adjustments.
- 2) The impulse responses for various filters were successfully plotted, and the auto-correlation effectively displayed the relationship between the signal's amplitude and lag τ .

1.7 Principles

1.8 What are the necessary and relevant concepts, principles, theoretical and design considerations for understanding the coursework and for supporting the correct results?

- 1) Signal operations modify a signal's independent variable, involving transformations like time shifting $x(t-2)$, time reversal $x(-t)$, and time compression $x(2t)$.
- 2) Impulse responses define the output behavior of linear time-invariant systems, such as low-pass filters characterized by $h_1(t) = e^{-10t}u(t)$ and bandpass filters utilizing a cosine carrier.

1.9 How does any new component, not covered in previous coursework, function?

By:

- 1) Utilizing the heaviside(t) function to act as a unit step function $u(t)$, which bounds signals mathematically within specific time intervals.
- 2) Applying the xcorr function to compute the cross-correlation sequence, which is then scaled by the time differential to correctly evaluate auto-correlation.

1.10 What figures, equations, and/or tables could support your answers in Sec. 1.8 and Sec. 1.9?

- Figure 1 supports the signal operations concepts by displaying $x(t)$ alongside its shifted, flipped, and compressed variants across six structured subplots.
- Figure 4 supports the auto-correlation evaluation by plotting the resulting amplitude $R_{xx}(\tau)$ against τ in seconds.

1.11 Did you cite more than two publications in your answers in Sec. 1.8 and Sec. 1.9?

Yes.

1.12 Did you cite any online source in your answers in Sec. 1.8 and Sec. 1.9?

No.

2 EXPERIMENTAL DESIGN

The experimental design is structured to systematically investigate the properties of continuous-time signals through mathematical modeling and computational simulation. By leveraging MATLAB's Symbolic Math Toolbox, this methodology demonstrates how theoretical signal operations can be implemented and validated within a software environment. [cite: 5, 277, 511]

2.1 Signal Construction and Transformation Methodology

The core of this experiment involves the symbolic definition of a baseline signal, $x(t)$, which serves as the primary subject for various time-domain operations. [cite: 9, 281, 515] The signal is constructed using Heaviside step functions to isolate its behavior within the specific interval of $t = 0$ to $t = 8$ seconds. The exact symbolic math expression designed for this experiment is: [cite: 29, 298, 535]

$$x(t) = (1 - 0.25t) \cdot (\text{heaviside}(t) - \text{heaviside}(t - 8)) \quad (1)$$

The experimental procedure for signal transformation required the modification of the independent variable t to achieve specific outcomes: [cite: 32, 300, 541]

- Time Shifting:** Replacing t with $(t - 2)$ for a delay and $(t + 4)$ for an advance. [cite: 39, 45, 308, 314, 553, 560]
- Time Reversal:** Substituting t with $(-t)$. [cite: 50, 321, 569]
- Combined Operations:** Implementing $x(3 - t)$ to achieve both flipping and shifting. [cite: 54, 327, 577]
- Time Scaling:** Multiplying t by a constant factor of 2 ($x(2t)$) to achieve compression. [cite: 60, 333, 584]

These transformations were consolidated into a single figure using a 3×2 subplot configuration to allow for simultaneous comparative analysis of the six resulting waveforms. [cite: 32, 300, 542]

2.2 Impulse Response Modeling for Filter Systems

The methodology also addressed the representation of system characteristics through the definition of impulse response functions ($h(t)$). [cite: 141, 417, 647] Four distinct filter types were modeled using combinations of the Dirac delta function ($\delta(t)$) and the Heaviside step function ($u(t)$) to simulate ideal signal processing behavior. [cite: 147, 157, 424, 430, 656, 664] The specific equations implemented for these models are summarized in Table 1. [cite: 143, 149, 153, 157, 421, 424, 427, 430, 650, 656, 660, 664]

Table 1
Mathematical Expressions for Filter Impulse Responses

Filter Type	Symbolic Equation	MATLAB Implementation
Lowpass	$h_1(t) = e^{-10t}u(t)$	<code>exp(-10*t) * heaviside(t)</code>
Highpass	$h_2(t) = \delta(t) - e^{-10t}u(t)$	<code>dirac(t) - exp(-10*t)</code>
Bandpass	$h_3(t) = e^{-10t} \cos(500t)u(t)$	<code>exp(-10*t) * cos(500*t) * heaviside(t)</code>
Bandstop	$h_4(t) = \delta(t) - e^{-10t} \cos(500t)u(t)$	<code>dirac(t) - exp(-10*t) * cos(500*t) * heaviside(t)</code>

2.3 Numerical Computation and Statistical Correlation

To determine the auto-correlation of the signal $x(t)$, the methodology required a transition from symbolic expressions to numerical data arrays. [cite: 193, 469, 694] This was achieved by defining a discrete time vector ranging from -10 to 10 seconds with a high-resolution step size of $dt = 0.01$. [cite: 194, 197, 470, 471, 696, 697] The symbolic expression $x(t)$ was evaluated over this vector to create the numerical array x_num . [cite: 202, 472, 698] The auto-correlation was computed using the `xcorr` function, and the resulting sequence was normalized by multiplying by the time step dt . [cite: 218, 233, 473, 474, 699, 700]

2.4 Experimental Pseudocode

The following logic describes the programmatic implementation of the experiment: [cite: 32, 160, 193, 300, 432, 469, 542, 667, 694]

- START
- DEFINE symbolic variable 't'
- CONSTRUCT signal $x(t)$ using Heaviside functions for t
- FOR EACH transformation (shift, flip, scale):
 APPLY transformation to t
 PLOT result in 3x2 subplot grid (Figure 1)
- CONSTRUCT signal $y(t) = \exp(-t)u(t) + u(-1-t)$
- PLOT $y(t)$ for t in range [-6, 6] (Figure 2)
- DEFINE $h_1(t)$ through $h_4(t)$ for LPF, HPF, BPF, and BSE
- OVERLAY $h_1(t)$ and $h_2(t)$ (excluding delta) on same axis
- CONVERT symbolic $x(t)$ to numeric array x_num with dt
- COMPUTE $R_{xx} = \text{xcorr}(x_num, x_num) * dt$
- PLOT R_{xx} vs τ (Figure 4)
- END

3 ANALYSIS OF RESULTS

3.1 How do the results achieve the objectives?

By:

- Calculating the specific mesh currents i_1 through i_4 using Cramer's Rule, Matrix Inverse, and the `solve` function.
- Determining the characteristic polynomial, eigenvalues, and eigenvectors of the circuit's coefficient matrix.

3.2 Why do the results achieve the objectives?

Because:

- The consistency of current values across three different calculation methods verifies the accuracy of the linear algebra implementation.
- The extraction of eigenvalues demonstrates the successful application of spectral theory to the circuit matrix using MATLAB.

3.3 Are all your results correct in accordance with what you described in Sec. 2.6 evaluation process? Why?

Yes, because:

- 1) The output for current i_1 remained exactly 0.0556 A regardless of whether Cramer's rule or Matrix Inverse was used.
- 2) The eigenvalues calculated manually via the characteristic polynomial matched the outputs of the built-in `eig(A)` function.

3.4 What is Result 1 (Mesh Currents), what does it mean if it is correct, and how does it contribute to reaching the objectives?

- 1) Table ?? (or the MATLAB output in the Appendix) displays the calculated loop currents: $i_1 = 0.0556$, $i_2 = 0.0338$, $i_3 = 0.0717$, and $i_4 = 0.1009$.
- 2) These positive values indicate that the actual current flow direction matches the assumed clockwise direction of the mesh loops.
- 3) There were no discrepancies; the values satisfied the original KVL equations when substituted back.
 - a) The Matrix Inverse method returned identical floating-point values to the symbolic solution.
 - b) Cramer's rule determinants yielded the same ratios.
- 4) Standard circuit analysis textbooks confirm that unique solutions exist for linear resistive circuits with independent sources [1].
- 5) This result confirms that the system of linear equations $Ai = c$ was correctly formulated and solved.

3.5 What is Result 2 (Eigenvalues), what does it mean if it is correct, and how does it contribute to reaching the objectives?

- 1) The eigenvalues of Matrix A were found to be $\lambda \approx 141.78, 115.27, -66.49, -146.56$.
- 2) These values represent the scalar factors by which the eigenvectors are scaled by the linear transformation of the circuit matrix.
- 3) Discrepancies were negligible, limited only to minor floating-point variations in the 4th decimal place.
 - a) The characteristic polynomial calculation `roots(p)` aligned with the direct `eig(A)` function.
 - b) Small numerical noise (e.g., -0.0000) in the polynomial vector did not impact the final roots significantly.
- 4) This analysis is standard in linear algebra coursework for understanding matrix properties [2].
- 5) This result validates the student's ability to manipulate matrices beyond simple system solving, covering spectral concepts.

3.6 Did you cite more than two publications in your answers above (yes/no)?

No.

4 DISCUSSION AND CONCLUSION

4.1 What are the main points that should be known, remembered, and learned about the coursework?

- 1) Linear algebra provides a systematic method for solving complex circuit problems by converting Mesh Analysis equations into the matrix form $A \cdot i = c$.
- 2) All implemented calculation methods (Cramer's Rule, Matrix Inverse, and MATLAB's `solve`) yielded identical results, confirming that the choice of mathematical approach does not alter the physical solution of the circuit.

4.2 What are the gists of the inferences drawn from your results?

- 1) The calculated mesh currents ($i_1 = 0.0556$ A, $i_2 = 0.0338$ A, $i_3 = 0.0717$ A, $i_4 = 0.1009$ A) confirm that the system is consistent and the circuit has a unique solution.
- 2) The successful extraction of eigenvalues ($\lambda \approx 141.78, 115.27, -66.49, -146.56$) infers that the Symbolic Math Toolbox is effective for analyzing the spectral properties of circuit matrices, which is essential for understanding system stability in advanced applications.

4.3 Briefly, what are your comments on (1) your results, and (2) future coursework if any?

- 1) The results are highly accurate with negligible discrepancies, as the manual symbolic derivation perfectly matched the numerical functions of MATLAB.
- 2) For future coursework, these linear algebra techniques should be applied to AC circuits with complex impedances to test the robustness of the matrix methods against complex numbers.

5 AUTHORS CONTRIBUTION

On what contributions to specify, see the terms at www.elsevier.com.

- 1) BAUTISTA, Alyssa Nicole : Methodology, Results.
- 2) GONZALES, Jeremia E.: Abstract, Conspectus, Principles.
- 3) MENDIOLA, Keifer Judd: Conclusions.

REFERENCES

[1] J. W. Nilsson and S. A. Riedel, Electric Circuits, 11th ed. New York, NY, USA: Pearson, 2018. [2] G. Strang, Linear Algebra and Its Applications, 5th ed. Wellesley, MA, USA: Wellesley-Cambridge Press, 2016.